

Summary & Highlights

Congratulations! You have completed this module. At this point, you know:

- Data pipelines move data from one place, or form, to another
- Data flows through pipelines as a series of data packets
- Latency and throughput are key design considerations for data pipelines
- Data pipeline processes include scheduling or triggering, monitoring, maintenance, and optimization
- Parallelization and I/O buffers can help mitigate bottlenecks
- Batch pipelines extract and operate on batches of data
- Batch processing applies when accuracy is critical, or the most recent data isn't required
- Streaming data pipelines ingest data packets one-by-one in rapid succession
- Streaming pipelines apply when the most current data is needed
- Examples of streaming data pipelines use cases, such as social media feeds, fraud detection, and real-time product pricing
- Modern data pipeline technologies include schema and transformation support, drag-and-drop GUIs, and security features
- Stream-processing technologies include Apache Kafka, IBM Streams, and SQLStream

Interactivity: Differentiate between Batch Processing and Stream Processing

Close X

In case you are having issues viewing the lab instructions below or prefer to view the instructions in a new browser tab, [click here](#)

**Skills
Network**

Can you differentiate between ETL and ELT?

Drag and drop the below terms into the correct bins

Batch Processing

- High Latency
- MapReduce
- Usually a scheduled activity
- Good for activities like payroll processing

Stream Processing

- Good for activities like weather sensor data processing
- Low latency
- Spark Streaming
- Realtime activity

Submit
Score 100 %
Start Over

1. What is the main purpose of a data pipeline?

- ☐ Storage
- ☒ Move data from one place or form to another

Correct, data pipelines move data from one place or form to another.

- ☐ Data container
- ☐ Loading

2. There are always stages that are bottlenecks in the data flow of pipelines. Which of the following is a simple way to load balance pipelines?

- ☐ Queues
- ☒ Parallelize

Correct, this distributes data packets as they arrive and helps balance loads.

- ☐ Serialize
- ☐ Buffers

4. Which of the following is the main reason to use Lambda architecture instead of Micro-batch, Streaming, or Batch data pipelines?

☒ Access to earlier data and speed is important

Correct, Lambda can be used in cases where access to earlier data is required but speed is also important.

☐ Help with load balancing

☐ Records are processed immediately

☐ Accuracy is critical

5. Which of the following data pipeline processes keep the pipeline running smoothly?

☐ Ingestion

☐ Monitoring

☒ Maintenance and Optimization

Correct, maintenance and optimization keep the pipeline up and running smoothly.

☐ Scheduling and Triggering

1. Which of the following data pipelines corresponds with the fraud detection use case?

1

- ☒ Streaming data pipeline
- ☐ Micro-batch data pipeline
- ☐ Lambda architectures
- ☐ Batch data pipeline

✓ **Correct**

Correct, streaming data pipelines are used for fraud detection.

2. Batch data pipelines usually run periodically on fixed schedules. Which of the following is another method to run these?

1

- ☒ Triggers
- ☐ Manually
- ☐ Flags
- ☐ Error occurrence

✓ **Correct**

Correct, Batch processes typically operate periodically on a fixed schedule – ranging from hours to weeks apart. They can also be initiated based on triggers, such as when the data accumulating at the source reaches a certain size.

Question 1

Which of the following data pipelines corresponds with the fraud detection use case?

✓ **Streaming data pipeline**

(Fraud detection requires real-time processing to identify and respond to suspicious activity immediately, which is best handled by a streaming data pipeline.)

Question 2

Batch data pipelines usually run periodically on fixed schedules. Which of the following is another method to run these?

✓ **Triggers**

(Batch pipelines can also be triggered by events, such as the arrival of new data or a manual trigger, in addition to fixed schedules.)

5. Which of the following data pipeline use cases is the simplest?

1

- ☒ File backup
- ☐ Send/receive messages
- ☐ Raw data preparation
- ☐ Transactional record movement

✓ **Correct**

Correct, the simplest pipeline is one which has no transformations and is used to copy data from one location to another, as in file backups.

Question 5

Which of the following data pipeline use cases is the simplest?

✓ **File backup**

(File backup is a straightforward use case involving copying data from one location to another without complex transformations.)

8. Which of the following pipeline monitoring considerations affects the amount of data that passes through the pipeline over time?

- ☒ Throughput
- ☐ Utilization
- ☐ Latency
- ☐ Logging and alerting system

✔ **Correct**

Correct, this is the volume of data passing through the pipeline over time.

9. Latency is the total time it takes for a single packet of data to pass through the pipeline. Which of the following limits latency?

- ☐ Bad data
- ☐ Small data packets
- ☒ Slowest process
- ☐ Data leak

✔ **Correct**

Correct, latency is limited by the slowest process in the pipeline.

Question 8

Which of the following pipeline monitoring considerations affects the amount of data that passes through the pipeline over time?

✔ **Throughput**

(Throughput measures the volume of data processed per unit of time, directly impacting pipeline capacity.)

Question 9

Latency is the total time it takes for a single packet of data to pass through the pipeline. Which of the following limits latency?

✔ **Slowest process**

(Latency is constrained by the slowest step in the pipeline, often referred to as the "bottleneck.")

10. Micro-batch data pipelines decrease the batch size. Which of the following do micro-batch pipelines **increase**?

- ☐ Storage
- ☐ Simple transformation
- ☐ Latency
- ☒ Batch process refresh rate

✔ **Correct**

Correct, the refresh rate of individual batch processes is increased to achieve near-real-time processing.

Question 10

Micro-batch data pipelines decrease the batch size. Which of the following do micro-batch pipelines increase?

✔ **Batch process refresh rate**

(Smaller batches mean more frequent processing, increasing the refresh rate but potentially adding latency compared to true streaming.)